RealPDE Track 2 (Long-Term Test-Time Adaptation): the final submission in detail

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Abstract

This document describes the final submission (Codabench submission 948041, archive `f11d2c1d`). It scored 82.452, first on the Track 2 development-phase leaderboard, and won Track 2 after the organisers' final evaluation, in which they retrained the top methods. The model is a packed ensemble of three sim-pretrained, real-fine-tuned and weight-souped SimVP-style networks, wrapped in a causal, gradient-free test-time pipeline. The pipeline has several parts: context normalisation, a per-trajectory online least-squares correction of the ensemble head, calibration of the fluctuation energy, damping of low-energy fluctuations, interval bounds chosen per trajectory to maximise the interval score, and a detector that resets the running statistics after an abrupt regime change. The whole step runs as CUDA graph replays scheduled over several GPU streams, using kernels that are bit-exact to the unfused computation. The submission is rebuilt from the released data by one command.

Contents

1 Task and scoring **2**

2 Data **2**

3 Backbone and training **2**

3.1 Architecture . . . 2
3.2 Causal context normalisation . . . 3
3.3 Loss . . . 3
3.4 Training stages . . . 3
3.5 Fitted auxiliaries . . . 3

4 The online pipeline (per step, all causal) **3**

4.1 Regime-change detector (REC) . . . 3
4.2 Online least-squares correction of the ensemble head (LSK) . . . 4
4.3 Energy calibration . . . 4
4.4 Prior and advection blend . . . 4
4.5 Low-energy fluctuation damping (LOWKE) . . . 4
4.6 Interval bounds . . . 4
4.7 Causality summary . . . 5

5 Execution **5**

5.1 CUDA graphs and fallbacks . . . 5
5.2 Kernel diets . . . 5
5.3 Multi-stream schedule . . . 5
5.4 Fused elementwise kernels . . . 5
5.5 Transport across the harness process boundary . . . 5

5.6 Resulting speed 6

6 Validation evidence for each component **6**

7 Result **6**

8 Reproducibility and environment **7**

9 Limitations **7**

1 Task and scoring

Each hidden trajectory of real PIV measurements behind a NACA4418 airfoil (angle of attack 0–20°, Reynolds number 3 000–28 000) is revealed window by window. At step t the model receives the normalised input window $x_t \in \mathbb{R}^{1\times 20\times 32\times 64\times 3}$ (channels u, v, p; p is zero and unscored on real data) and the ground truth y_{t-1} of the previous step. It returns a prediction $\hat{y}_t$ and, optionally, per-element bounds $[\ell_t, h_t]$. The input and target windows overlap in stride 20, so window k's target is window $k{+}1$'s input. Everything inside `ttt_step` and `reset_ttt_state` is timed.

Five subscores are combined into the final score; the combination is not published. Our fit to the board is $\approx 0.48\,\mathrm{rel}_{L2} + 0.07\,\mathrm{TKE} + 0.10\,\mathrm{MVPE} + 0.09\,\mathrm{time} + 0.28\,\mathrm{SPS}$. The Safe Prediction Score is computed per element as

$$\mathrm{elem} = \big[0.5(1-p_{\mathrm{dm}}) + 0.3(1-p_{\mathrm{tke}}) + 0.2(1-p_{\mathrm{mvpe}})\big]\, e^{-(h-\ell)/\sigma_g}\, \mathbb{1}[\ell \le y \le h], \qquad p_x = \frac{e_x}{0.5 + e_x}, \quad (1)$$

where e_x are the per-window relative errors of the prediction. The per-window TKE error therefore enters both the TKE subscore and the SPS accuracy factor. The time subscore is $100/(1+\sqrt{t/0.72896\,\mathrm{s}})$.

2 Data

Only the released `train_sim` (100 simulation files) and `train_real` (82 files) are used; no external data or pretrained weights. `train_real/7575_0.h5` duplicates another file (organizer announcement) and is excluded at every stage, which leaves 81 real files. Simulation data are subsampled to the real 32×64 grid and rescaled to physical units, and p is masked, so that pretraining sees exactly the real input layout. The backbone uses no augmentation; the energy-net fits use an on-the-fly flip in y.

Development decisions used leave-conditions-out (LCO) folds that hold out whole angle-of-attack / Reynolds regimes:

- **fold1:** used for selection.
- **fold0:** the primary held-out fold.
- **foldR:** held-out Reynolds numbers.
- **Rx:** Reynolds extrapolation.
- **foldS-lo, foldS-hi:** simulation-only regimes, as beyond-range surrogates.
- **27 stress sets:** regime switches, velocity scaling, PIV dropout, noise, long trajectories and others.

The private test contains unseen parameter regimes, so every component below also had to be non-negative on the private-regime proxies (foldR, Rx and the foldS sets).

3 Backbone and training

3.1 Architecture

The backbone is a SimVP-style network with gated spatiotemporal-attention blocks (`b5_h192_blk6_rf11`): encoder width 128, hidden width 192, 6 gSTA blocks and receptive field 11, 3.39 M parameters in total.

3.2 Causal context normalisation

The backbone is wrapped in causal context normalisation, which is the core test-time adaptation. For each trajectory the wrapper keeps a per-pixel running mean and standard deviation of every frame revealed so far: input frames and previously revealed targets, never the current target. These two maps are fed to the network as extra channels, and the network's output is de-standardised with them. This adapts the operating point to the trajectory's own regime with no gradient step. The state is cleared in `reset_ttt_state`.

3.3 Loss

The loss is a bounded Welsch (correntropy) loss, mixed 75/25 between physical and per-channel-standardised space, with normalisers fixed from the first batch of each training stage.

3.4 Training stages

1. **Sim pretraining.** All 100 simulation files. AdamW, learning rate $5 \cdot 10^{-4}$, no weight decay, batch size 8, 3 % warmup and then cosine decay over a 20k-iteration schedule. The run is snapshotted at iteration 10k, a length selected on fold1.
2. **Real fine-tuning.** 8k iterations on the 81 real files with the same optimiser, starting from the sim snapshot.
3. **Weight soup.** Four fine-tunes (seeds 0–3) from the same snapshot are averaged uniformly in weight space (computed in fp64, cast to fp32). Because the members share one loss basin, the soup is better than any single member at no extra inference cost.
4. **Ensemble ($K = 3$).** Stages 1–3 are repeated with sim-pretrain seeds 0, 1 and 2. Diversity comes from the pretrain seed; souping across seeds is invalid because the members are in different basins. The three soups are packed into one batched network (10.2 M parameters): stem concatenation, grouped convolutions and batched 1×1 convolutions. The output is the members' mean in standardised space.
5. **fp16 core.** Everything inside the packed backbone runs in half precision; the surrounding pipeline stays fp32. The observed headroom to fp16 overflow is at least 1 800×, and the composite changes by at most $|\Delta| \leq 0.0002$.

3.5 Fitted auxiliaries

- **Prior:** a small per-condition seed that stabilises the first windows, with decaying influence.
- **Energy net ("KE3"):** four energy nets (seeds 0–3) predict the fluctuation energy of the coming window from causal features. Their mean is written exactly as one dense network of 4× width (block-diagonal weights; fp64 check $\leq 5 \cdot 10^{-9}$), so it runs as a single net on tensor cores.
- **Interval-width net ("W2"):** a small CNN that predicts the log-ratio of the local error scale to the pipeline's running error scale. Its inputs are causal features, the lead index and the previous window's error map (EMA). It is fitted only on out-of-sample residuals from 4-group cross-fitting over the 81 files.

4 The online pipeline (per step, all causal)

Every quantity below uses only x_t, the revealed $y_{<t}$ and fixed constants. Grids, thresholds and constants are literals in `submission.py`; the retrain chain fits none of them. The stages run in the order listed.

4.1 Regime-change detector (REC)

The detector computes $d = |\log(\bar{U}_{\text{win}}/\bar{U}_{\text{run}})|$ from the mean physical u of the current input frames and the running mean since the last reset. If $d > \log 1.5$, the context-normalisation accumulator, the error, centre and blend moments, and the least-squares moments of Section 4.2 restart. Mean-type statistics are rescaled by $\bar{U}_{\text{win}}/\bar{U}_{\text{run}}$ and given a pseudo-count of 6.

On all 81 real files the statistic stays below 0.085, so the detector never fires and REC is bit-identical to its absence on normal streams. On synthetic regime-switch streams it recovers within 0–2 windows.

4.2 Online least-squares correction of the ensemble head (LSK)

The packed head computes $\frac{1}{K}\sum_k (W_k h_k + b_k) = W_{\text{cat}} H$ with $W_{\text{cat}} = [W_1 \dots W_K]/K$. Let V hold the top-8 right singular vectors of the u, v rows of W_{cat}; they are computed at model construction from the trained weights, so no extra fit is needed. The feature vector is $\phi = [V^\top H; 1]$, and s is the causal context standard deviation. For each channel, from the revealed error $e = y_{t-1} - \hat{y}^{\text{raw}}_{t-1}$:

$$\Phi \leftarrow \rho\Phi + \textstyle\sum s^2 \phi\phi^\top, \qquad B \leftarrow \rho B + e\,(s\phi)^\top, \qquad M = B(\Phi + \lambda I)^{-1}, \tag{2}$$

with $\rho = 0.97$ and $\lambda = 30\,\mathrm{tr}(\Phi_1)/9$ (fixed from the first revealed window). The raw u, v output of all 20 leads is then corrected by $(M\phi)s$ before any downstream stage.

The moments are held in fp64 and the 9×9 solve uses `solve_ex` with no host synchronisation. A non-finite guard skips bad pairs. The moment update runs on a side GPU stream, so only one GEMM and the apply sit on the step's critical path. A build-time self-test uses the configured ρ and falls back to "LSK off" on any mismatch. REC multiplies the moments by its keep factor.

4.3 Energy calibration

The predicted fluctuation amplitude is rescaled so that its energy matches a blend of the energy-net prediction and the running mean of revealed energy. The rescale adds the mean back so that variance is preserved, and the gain is clamped to $[1/20,\ 20]$.

The blend weight $b \in \{0.45, 0.60, 0.75, 0.85, 0.925\}$ is chosen per trajectory. Every candidate is computed each step, batched, and the one with the best EMA ($\rho = 0.95$) of the previous window's official composite is emitted. Before any evidence exists, $b = 0.85$.

4.4 Prior and advection blend

The prediction is blended with an advected copy of the most recently revealed frame. The ridge strength of this blend's least-squares fit is chosen per trajectory from $\{0.125, 0.25, 0.5, 1\}$ by the same previous-window-score rule.

4.5 Low-energy fluctuation damping (LOWKE)

Let m_t be the 20-frame mean of the prediction and $K_p = \frac{1}{2}(\mathrm{var}_t u + \mathrm{var}_t v)$ its energy map. For windows ≥ 2,

$$g = \mathrm{clamp}\Big(\big(K_p/(1.5\,\overline{K_p})\big)^{0.25}, 0, 1\Big), \qquad \hat{y} \leftarrow m_t + g\,(\hat{y} - m_t). \tag{3}$$

This removes weakly correlated fluctuations where the model itself predicts calm flow. rel_{L2} and SPS rise and TKE drops slightly. The bounds and internal state are untouched.

4.6 Interval bounds

- **Width:** the W2 half-width, multiplied by the running error scale and by an adaptive multiplier $\kappa \in \{0.85, 0.925, 1, 1.075, 1.15\}$. κ is chosen separately for leads 1–2 and leads 3–20 by the EMA of the exact SPS element score on the previous window.
- **Centre:** the centre is shifted toward the raw ensemble output, $c' = c + w(\hat{y}^{\text{raw}} - c)$. The weight $w \in \{0.30, 0.45, 0.60, 0.75, 0.90\}$ is chosen per trajectory by the same rule. The fallback is $w = 0.75$, and window 0 uses $w = 1$.
- **Scored mask:** the per-lead error maps use the scorer's exact scored mask, $|y| > 10^{-6}$ on the normalised target.

Bounds are returned on every step of every run, as the all-or-nothing rule requires.

4.7 Causality summary

Stage	Information used
Context normalisation, REC	frames revealed up to $t-1$; the current step's input frames
LSK	revealed errors up to $t-1$; the current step's own head features
Energy / blend / ridge / κ / w selectors	scores of previous windows only (EMA)
Prior, W2, energy nets	fixed after training (training files only)
Current target y_t	never used for step t

5 Execution

5.1 CUDA graphs and fallbacks

From window 2 onward the entire step is one CUDA-graph replay. Windows 0 and 1 have their own captured graphs, and every graph has a bit-exact eager fallback. Each graph is self-tested against eager execution at model construction, and any failure degrades gracefully. An RNG probe guards against capture-time corruption of the CUDA random state.

5.2 Kernel diets

All diets preserve outputs exactly, except KD5, which changes rounding order only ($|\Delta\text{composite}| \leq 0.0004$).

- **Fused selectors:** the candidate scoring is fused into fewer kernels.
- **Leaner step body:** the backbone and "mega" stage are rewritten with fewer kernels.
- **KD5:** near-exact convolution rewrites.
- **First-two-window graphs:** windows 0 and 1 are captured with a side stream and private memory pools.

5.3 Multi-stream schedule

The step graph is captured with fork/join edges. Work that does not depend on the backbone output (error and loss bookkeeping, running variances, width-net updates, adaptive selectors, input-only preparation) runs on parallel streams alongside the backbone. The LSK update and the LOWKE gain also have their own streams. Every cross-stream read-after-write and write-after-read hazard is ordered by an event.

This was verified in three ways: a race checker using transitive happens-before (0 conflicts); replays with deliberate delays that invert each fork/join completion order; and runs under foreign GPU load and different connection limits. All comparisons matched bit for bit.

5.4 Fused elementwise kernels

The longest elementwise chains on the critical path are fused into `torch.cuda.jiterator` kernels. They are written with round-to-nearest intrinsics (`__fmul_rn`, `__fadd_rn`, ...) so that NVRTC cannot contract them into fused multiply-adds, which keeps them bit-identical to the unfused graph. They are compiled and bit-tested at construction, outside the timed region; any failure falls back.

5.5 Transport across the harness process boundary

The official harness runs the model in a worker subprocess.

- **By-value outputs (BVR):** outputs are returned as a zero-copy tensor subclass whose pickled form is its values, so the parent receives ordinary tensors with identical values. This proved decisive on the board.

- **Other transport paths:** fd-shared single-storage outputs, a listener warm-up during construction (untimed), and direct CUDA I/O cover the other possible harness designs. They are inert where they do not apply.

5.6 Resulting speed

The eval harness reports 4.84 ms per step. Of that, about 2.6 ms is our GPU work on the A800-class device; the rest is harness transport.

6 Validation evidence for each component

Each component was adopted under a gate declared in advance: selection on fold1, then a single held-out evaluation on fold0, foldR, Rx and the foldS sets. Most were also reviewed by an independent Fable reviewer and verified on the exact archive: bit-identity checks, out-of-process harness emulation, fuzzing, reset probes and determinism.

The table lists the change in the local composite (time held fixed) at the moment each component was added to the stack of that time. It therefore shows increments, not a clean ablation of the final stack. Board transfer is about 0.6× fold0.

Component	fold1	fold0	foldR	Rx	foldS-lo	foldS-hi
Sim pretraining	+0.105	+0.184	+0.064	+0.177	+0.186	−0.181
Soup of 4	+0.022	+0.075	+0.048	+0.041	+0.024	−0.001
2nd ensemble member	+0.080	+0.091	+0.082[a]	+0.060[a]	+0.073[a]	+0.084[a]
3rd member + fp16 core	+0.048	+0.039	–	–	+0.026	−0.015
Centre blend	+0.110	+0.089	+0.114	–	–	–
Adaptive centre weight	+0.001	+0.001	–	–	+0.002	+0.055
Adaptive width / energy blend	+0.017	+0.027	+0.011	–	+0.024	+0.046
Gain clamp 20	+0.006	+0.008	+0.031	+0.019	−0.004	+0.000
Exact scored mask	+0.003	+0.005	+0.004	+0.009	−0.001	+0.005
Beyond-range hedge (AXE)	+0.000	+0.009	–	–	+0.009	+0.235
4-seed energy net (KE3)	+0.018	+0.043	+0.020	–	+0.057	+0.115
LOWKE (net of time)	+0.022	+0.033	+0.021	–	+0.037	+0.035
LSK ($\lambda_{\text{rel}} = 30$)	+0.018	+0.041	+0.027	+0.034	–	–
LSK ρ 0.9→0.97[b]	+0.004	+0.010	+0.005	+0.007	+0.008	+0.033

[a] measured on an independent 4090 build. [b] Missed the fold0 bar of +0.015 declared in advance. It was adopted as a disclosed override because it was non-negative on every held-out and private-proxy set, and because an independent refit on the 4090 reproduced it (fold0 +0.009, 5/5 sets ≥ 0).

Notes on the table:

- REC is bit-identical on every normal stream and gains +17–22 composite on a regime-switch file.
- Resetting the LS moments on REC is bit-identical off-switch and gains +0.036 on the switch set.
- The timing layers are output-identical except KD5 (rounding order only). On a 3090 they save: fused selectors −0.11 ms, leaner step body −0.11 ms, KD5 −0.115 ms, first-two-window graphs −0.05 ms, multi-stream schedule −0.42 ms (−0.50 ms on an A100), fused kernels −0.12 ms and the final critical-path cuts −0.07 ms.
- By-value transport saves about 0.7 ms per step on the board.

In a 27-set stress battery with unseen-regime surrogates, the pipeline beat the best trivial predictor on every set, with no exceptions and no non-finite outputs.

7 Result

	rel_l2	TKE	MVPE	time	SPS	final
Development leaderboard, 27 Sep 2026	94.939	81.017	94.500	92.464	45.211	**82.452**

The pre-registered prediction was 82.455 (82.40–82.51). The time subscore corresponds to 4.84 ms per step. This was first on the development-phase leaderboard (Codabench submission 948041). In the organisers' final evaluation, for which they retrained the top methods, this submission won Track 2.

8 Reproducibility and environment

- **One command:** `bash scripts/retrain_all.sh`, run in `final_submission/realpde/` of the code release, rebuilds `submission.zip` from the released data. The chain has 13 idempotent steps and uses seed 0 throughout.
- **Wall time:** about 2–3 h on an RTX 3090 (2 h 47 measured from a fresh clone), about 1 h on an RTX 4090 and about 2 h 15 on an A100 with the evaluation software versions. A from-scratch A100 retrain scored within ±0.03 of our build on every local subscore.
- **What is reproduced exactly:** code entries are byte-identical, and on the same GPU so are `model.pth` and `prior.npz`. Under GPU contention the sim-pretraining snapshot can differ bitwise, so the correctness bar is score parity, not byte identity.
- **Runtime:** Python 3.10, PyTorch 2.2.2 (CUDA 12.1), numpy 1.26. The NVRTC shipped with torch is needed for the fused kernels, which fall back if it is missing. Inference is offline and reads nothing from disk beyond the archive; the extracted archive is about 60 MB.
- **Robustness of the chain:** an opt-in serial mode supports clusters that allow one process per GPU. Reused artefacts are validated before a step is skipped. An open-file audit confirmed that only the released training data are read.

9 Limitations

- TKE and the remaining SPS gap are limited by the backbone's forecast correlation at late leads and high Reynolds numbers. Output-side corrections tested on top of the final stack (spectral, temporal and physical projections; phase warps; per-trajectory reduced models) did not improve it.
- The highest-Reynolds 0° files remain the weakest regime: LSK is slightly negative on a few of them.
- Synthetic bursts of all-zero frames degrade the LS correction. No released real frame has this property.
- Timing levers that depend on the harness were validated on emulations and on the board; the exact private harness is not public.